

POSS functionalized graphene oxide nanosheets with multiple reaction sites improve the friction and wear properties of polyamide 6

Lei Sun, Ming Cao, Fan Xiao, Jiaopeng Xu, Yingbo Chen^{*}

School of Materials Science and Engineering, Municipal Key Laboratory of Advanced Fiber and Energy Storage Technology, Tiangong University, Tianjin, 300387, PR China

ARTICLE INFO

Keywords:

Polyamide 6
Graphene oxide
POSS
Friction and wear

ABSTRACT

Frictional and wear properties of polyamide 6 (PA6) are very important for its use as an engineering plastic. In this study, octaamino-polyhedral oligomeric silsesquioxane (POSS) functionalized graphene oxide (GO) was used to fabricate a PA6/reduced GO-POSS (rGO-POSS) nanocomposites. Compared with PA6, the PA6/rGO-POSS nanocomposites with 0.3 wt% GO-POSS has a decrease of friction coefficient by 86.36% from 0.44 to 0.06. The depth and width of the wear scar were reduced by 79.64% and 70.69%, respectively. Meanwhile, the wear rate reached the lowest and dropped by about 65.72%. The research found that GO-POSS is self-fixed in the matrix and effectively increases the interface bonding strength.

1. Introduction

As an engineering plastic, polyamide 6 (PA6) not only has the characteristics of low density, friction resistance, corrosion resistance, and stable chemical properties [1,2], but also has the advantages of low cost, simple manufacturing process, and strong designability for products manufacturing. It has been successfully used in aerospace, electronic appliances, automotive industry [3] and other fields. PA6 has excellent tribological properties and mechanical properties [4], which is mainly due to the strong intermolecular forces [5] by forming hydrogen bonds among the amine groups in its main chain. However, PA6 has limited mechanical properties, unstable dimensions and high water absorption. A lot of heat is generated during the friction process, and the polymer has poor mechanical stability under higher temperature conditions [6]. The energy consumption caused by friction and the failure of PA6 are mainly due to abrasive wear and adhesive wear [7–12], hence, it is of great practical significance to solve the above problems and develop high-performance wear-resistant products [13–15].

As two-dimensional materials, graphene (Gr) has already attracted attention owing to its excellent electrical properties, thermal properties, mechanical properties and wear resistance [16–21]. The Hummers' method is frequently used to strip and oxidize Gr to obtain graphene oxide (GO) [22,23]. GO is more suitable for covalent functionalization than Gr. What is more important is that its surface has ample oxygen-containing functional groups, which could carry out more

susceptible to bonding and chemical reactions with other functional groups [24]. Currently, starting from the oxygen-containing groups on GO nanosheets, a series of modified GO-based nanocomposites have been obtained by grafting various polymers or some other nanomaterials through chemical bonding methods. For example, Chen et al. [25] reported that amino-modified GO was embellished by N-(2-aminoethyl)-3-aminopropyltrimethoxysilane (AAPS), the polymer-modified GO has great dispersion and high thermal stability, prominently reducing the friction coefficient of the nanocomposite material. Lee et al. [26,27] reported that 2-bromo-2-methylpropionic acid (BMPB) immobilized on the surface of GO by atom transfer radical polymerization, which was used for initiating the polymerization of esters and butyl acrylate that improve the solution processing properties of this type of material. Wang et al. [28] used biomimetic dopamine (PDA) as a bridging agent to graft a continuous and dense GO layer on the surface of basalt fiber (BF). The introduction of GO makes the interface bonding property of BF and PA6 substrate with greater improvements. Meanwhile, the mechanical and tribological performance of PA6 has been markedly improved. Therefore, based on the above elaboration, GO nanosheets has outstanding properties that are suitable for modification of PA6 with great wear resistance.

Polyhedral oligomeric silsesquioxane (POSS) is an inorganic core composed of Si–O alternately connected silicon-oxygen frameworks. Its shape is like a “cage” with a three-dimensional size between 1 and 3 nm. The large number of molecules in the unit volume of POSS leads to high

^{*} Corresponding author.

E-mail address: chenyingbo@tiangong.edu.cn (Y. Chen).

<https://doi.org/10.1016/j.triboint.2020.106747>

Received 7 September 2020; Received in revised form 20 October 2020; Accepted 26 October 2020

Available online 4 November 2020

0301-679X/© 2020 Elsevier Ltd. All rights reserved.

molecular polarization, meanwhile, the Si–O alternately connected silicon-oxygen skeleton can inhibit the chain movement of polymer molecules, thereby giving the material better dielectric properties, optical properties, thermal stability and mechanical properties [29,30]. In terms of toughness, POSS nanoparticles can stop the development of micro-crack tips, and can trigger cracks or shear bands or molecular chains to rearrange. The elasticity of the “cage” can act like a “marble”. Also, the application of POSS in polymers mainly depends on the reactive group (R), as shown in Fig. 1, which can be grafted or polymerized with the polymer through the reactive R group, resulting in chemical bonding between the polymers [31]. Besides, the diversity of organic groups in the siloxane cage provides a larger binding site for controlling the microstructure of POSS-based hybrid materials [32–34]. For example, Liao et al. [35] used trisilanol-phenyl-cage POSS to modify the chemically stripped molybdenum disulfide (MoS_2) nanosheets, the resulting POSS- MoS_2 nanocomposites could be well dispersed in poly-methyl methacrylate matrix. Zhou et al. [36] compounded the PA6/POSS nanocomposites with the aid of an applied electric field using phase inversion method. Under an applied electric field, the polymer crystallinity and hydrophilicity remain stable with the strength of the electric field increased. Hence, as a category of nanomaterial with excellent solubility, strong thermal stability and excellent mechanical properties [37–39], POSS can be used as a “tentacle” for functionalization of GO nanosheets.

In this study, the octopus-shaped GO-POSS nanosheets were prepared by covalent reaction of amino group in POSS and carboxyl group in GO, and GO-POSS was grafted to PA6 chain end through in-situ polymerization of caprolactam. PA6/reduced GO-POSS (rGO-POSS) nanocomposite was fabricated by the reaction of amino groups in the end of the PA6 chain and carboxyl groups in GO-POSS, companying with the reduction of oxygen containing groups in GO. The structures and mechanical property, as well as friction and wear resistance of the PA6/rGO-POSS nanocomposite were studied.

2. Experimental section

2.1. Chemicals and materials

Caprolactam (CL, 99%) and 6-aminocaproic acid (99%) were directly purchased from Shanghai Aladdin Biochemical Technology Co., Ltd., China. Octaamino POSS (95%) was obtained from Hybrid Plastic Co., Ltd., USA. Graphene oxide powder was supplied by Suzhou Carbon-Rich Graphene Technology Co., Ltd., China. Anhydrous methanol (99.5%) was bought from Tianjin Guangfu Fine Chemical Research Institute Co., Ltd., China. Formic acid (88%) was purchased from Tianjin Yuyuan Chemistry Co., Ltd. All chemicals were used directly without

purification.

2.2. Preparation

2.2.1. Preparation of GO-POSS

0.01 g dried POSS (50 °C, 24 h in vacuum oven) was dispersed in 30 mL methanol by magnetic stirring and ultrasonic dispersion. 0.1 g GO was added to 50 mL methanol, and the solution was ultrasonically dispersed for 60 min to obtain a clear light brown solution. Then, the former solution was poured into the latter. The mixed solution was reacted at 60 °C for 8 h under magnetic stirring. After the reaction, the solution was centrifuged with a speed of 8000 rpm, the product was dispersed repeatedly in deionized water and centrifuged for 6 times to remove unreacted POSS. The final product was dried at 60 °C for 24 h in the oven before using.

2.2.2. In-situ polymerization synthesis of PA6/rGO-POSS

CL and 6-aminocaproic acid were put in an oven for 24 h at 60 °C to remove excess water. A typical procedure [40–44] to prepare PA6/rGO-POSS nanocomposites was implemented as follows: 27 g CL, 3 g 6-aminocaproic acid and GO-POSS (according to the proportion of 0.1%, 0.3%, 0.5%, 0.7%, 0.9%) were added in a four-necked flask with ultrasonic at 80 °C for 30 min. The mixture was heated under nitrogen atmosphere with steady mechanical stirring according to the following procedure: (1) keep the temperature at 80 °C for 10 min to remove adsorbed water, (2) rise up the temperature to 220 °C with a rate of 10 °C in every 5 min and keep the temperature constant for 1.5 h, (3) increase the temperature to 260 °C with a rate of 10 °C in every 5 min and keep the temperature constant for 4 h. The synthetic polymer nanocomposite is named as PA6/rGO-POSS (since the temperature has reached the thermal reduction temperature of GO, some GO is reduced). As a comparison, GO and GO-POSS were reduced to rGO and rGO-POSS under the similar heating procedure without adding CL and 6-aminocaproic acid.

2.3. Characterization

2.3.1. Fourier-transform infrared spectrometer (FTIR) analysis

The chemical structure of GO, rGO, GO-POSS, rGO-POSS and g-PA6/rGO-POSS was characterized by a Nicolet iS50 FTIR (German). The temperature is 15–30 °C and the relative humidity is below 65%. Each specimen was dried in oven (60 °C, 8 h) before test.

2.3.2. Raman spectra analysis

The chemical structure analysis of GO and GO-POSS was characterized by the XploRA PLUS laser confocal scanning phase Raman

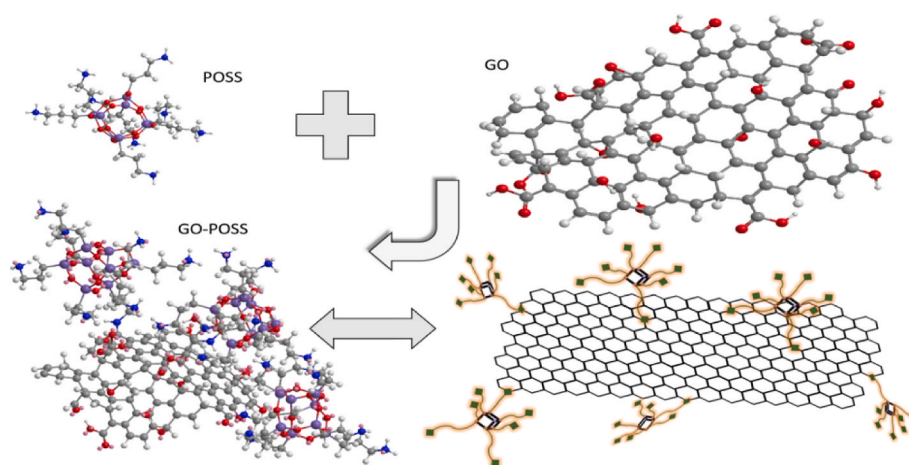


Fig. 1. Molecular simulation of GO-POSS synthesis process.

spectrometer of Horiba company (Japan). Raman spectra were recorded from 500 to 2500 cm^{-1} using a 532 nm argon ion laser.

2.3.3. X-ray photoelectron spectroscopy (XPS) analysis

The XPS analysis of GO, rGO, GO-POSS and rGO-POSS was performed on a D8-ADVANCEX, BRUKER Co. (German). The scanning range is 5° – 50° .

2.3.4. Thermal conductivity analysis

DRL-III thermal conductivity tester (China) was used to test the thermal conductivity. The heat flow method is used for test analysis. The temperature of the hot side is 60°C and the cold side is 20°C with a load of 150 N and a thickness of 1.6 mm. The heat transfer area is 314 mm^2 . Each specimen was dried in oven at 60°C for 8 h, and three samples were used for testing.

2.3.5. Differential scanning calorimetry (DSC)

DSC was used to characterize the thermal stability and crystal form of nanocomposites. It was characterized by a DSC200F3 from NETZSCH Co. (German) under nitrogen atmosphere with the heating rate of $5^\circ\text{C}/\text{min}$. The calculation formula of crystallinity (X%) is according to Equation (1) [45]:

$$X = \Delta H_m / \Delta H_m^* \quad (1)$$

Where ΔH_m is melting enthalpy of the nanocomposites, and ΔH_m^* (221.42 J g^{-1}) is the melting enthalpy of 100% crystalline PA6. Each specimen was dried in oven (60°C , 24 h) before test.

2.3.6. X-ray diffraction (XRD) analysis

XRD of GO and GO-POSS was performed with a D8-DISCOVER X-ray diffractometer (German) using Cu K α radiation ($k = 0.154056\text{ nm}$). XRD data were collected from 5° to 50° .

2.3.7. Atomic force microscopy (AFM) analysis

The morphologies of GO and GO-POSS were examined by AFM. It was operated by an Icon, BRUKER Co. (German) in a tapping mode. The spin coating method was used to evenly coat samples on the surface of the silicon wafer and then allowed to dry at room temperature.

2.3.8. Scanning electron microscope (SEM)

Phenom XL SEM (Netherlands) was used to observe the surface morphology and fracture morphology of PA6/rGO-POSS nanocomposites. The cross section morphology of the nanocomposites was characterized via Thermal Field-Emission Scanning Electron Microscope (SEM Gemini SEM500, German). Each specimen was dried in oven (60°C , 4 h) before test.

2.3.9. Mechanical properties

The tensile performance of PA6/rGO-POSS nanocomposites was measured with a FKV1000 from AGAINSO Co. (China), according to the national standard (GB/T1040-1992). The tensile test speed is $10\text{ mm}/\text{min}$ with the test fixture spacing is $30 \pm 0.5\text{ mm}$, and five samples were used for testing. LX digital display Shore D hardness tester (China) was used to test the hardness of nanocomposites. The samples were dried in an oven at 60°C for 4 h, and five points were measured for each sample.

2.3.10. True color confocal microscope

The worn surface topographies of PA6 nanocomposites were performed with a Zeiss CSM700 from ZEISS Co. (German). Each sample was sonicated with ethanol for 1 h before testing, and then dried in an oven at 60°C for 4 h.

2.3.11. Friction and wear test

The friction and wear performance of PA6 nanocomposites was tested under dry sliding by using a UMT-2 from Tianjin Research

Institute for Advanced Equipment, Tsinghua University. The block specimens were in size of $20\text{ mm} \times 20\text{ mm} \times 1.8\text{ mm}$, and its ball-on-block reciprocating motion was carried out on the 45# steel ball in point touch at a load of 12 N and sliding frequency of 4 Hz, lasting for 2 h by 5 mm friction stroke. The wear surface morphology of the PA6 nanocomposite was checked by a Zeiss CSM700 from ZEISS Co. (German). Obtain the loss mass (Δm , mg) by weighing method and then find the wear volume loss (ΔV , mm^3). The wear rate (K , $\text{mm}^3\text{N}^{-1}\text{m}^{-1}$) was calculated according to Equation (2):

$$K = \Delta V / FtL \quad (2)$$

where F stands for applied load (N), t stands for experimental duration (min), and L stands for stroke length in one cycle (m). Each sample was sonicated with ethanol for 1 h and dried in an oven at 60°C for 4 h before testing.

3. Results and discussion

3.1. Characterization of GO, GO-POSS and PA6/rGO-POSS

As shown in Fig. 1, the synthesis of GO-POSS nanomaterials involves the amide reaction of the carboxyl group of GO and the amine group of POSS. The final product synthesized in this way has an “octopus” structure, in which cage-shaped POSS can be used as a “tentacle”, and GO nanosheets can be regarded as a “head”. Herein, POSS was grafted onto the GO nanosheet and occupied a reactive site grafted with PA6. However, the amino groups on POSS can also be regarded as sites for reaction with PA6, thereby increasing the grafting sites for the reaction between GO-POSS and PA6. This octopus structure is grafted with PA6 by in-situ polymerization and then self-fixed in the PA6 matrix. The formed compact octopus-shaped structure can not only effectively transfer stress, but also prevent the initiation and propagation of cracks.

In order to investigate whether POSS and rGO-POSS were grafted onto GO and PA6, respectively, FTIR spectroscopy analysis was performed on the GO, POSS, GO-POSS, rGO, rGO-POSS and purified g-PA6/rGO-POSS. This purified g-PA6/rGO-POSS is to dissolve PA6/rGO-POSS in formic acid and separate ungrafted PA6 by segmented centrifugation. In Fig. 2(a), the major absorption bands of GO-POSS at approximately 1116 cm^{-1} , 1630 cm^{-1} , 1732 cm^{-1} , 3421 cm^{-1} , are corresponding to the Si–O–Si stretching absorption, the –NH–C(O)– stretching vibration [46], the –C(O)–OH– stretching vibration, and the –OH stretching vibration, respectively. Besides, compared with the Si–O–Si stretching vibration peak of POSS at 1088 cm^{-1} , the Si–O–Si (1116 cm^{-1}) [31] stretching vibration peak of GO-POSS has a red shift, which exactly proves that GO reacts with POSS. Compared with rGO-POSS, g-PA6/rGO-POSS appears two new vibration absorption peaks –C–N– and –NH (terminal amino group) of PA6, are corresponding to 1533 cm^{-1} and 1360 cm^{-1} . Meanwhile, taking the Si–O–Si stretching vibration peak of rGO-POSS at 1116 cm^{-1} as a comparison, the Si–O–Si of g-PA6/rGO-POSS has a blue shift from 1116 cm^{-1} to 1056 cm^{-1} [31]. The above statement decidedly proves that rGO-POSS and PA6 have been grafted through amide bond. In Fig. 2(b), we compared the dispersion of PA6/rGO and PA6/rGO-POSS nanocomposites in formic acid. It could be clearly seen that there are a large number of suspended particles in the PA6/rGO dispersion, which is not observed in the PA6/rGO-POSS dispersion. And that the color of PA6/rGO is lighter than that of the PA6/rGO-POSS dispersion at the beginning. After three days, most of the fine particles in the PA6/rGO dispersion were settled down to the bottom and the color of the dispersion became light. On the contrary, the PA6/rGO-POSS dispersion did not change much after three days, and the color remained unchanged.

The XRD patterns of GO and GO-POSS samples were illustrated in Fig. 3(a). The characteristic peak of GO was observed at $2\theta = 9.8^\circ$, corresponding to the layer spacing of 0.90 nm. However, the highest peak intensity of GO-POSS appears at 10.1° , which corresponds to the

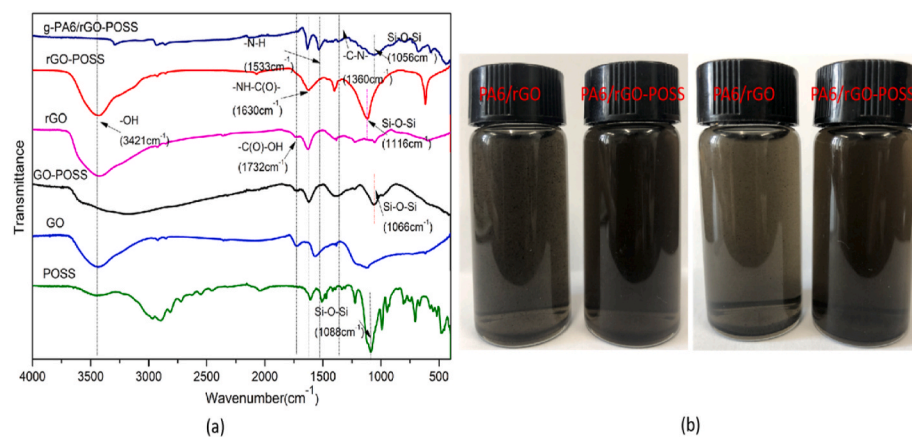


Fig. 2. (a) Fourier-transform infrared of GO, rGO, GO-POSS, rGO-POSS and g-PA6/rGO-POSS. (b) Three days before (left picture) and after (right picture) for 0.1 wt % nanocomposites dispersed in formic acid (0.05 g/mL).

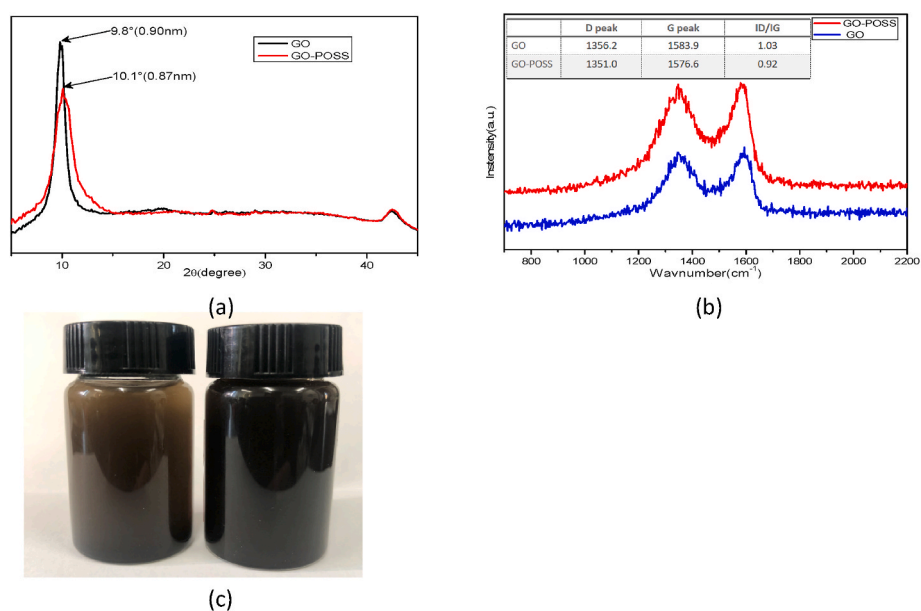


Fig. 3. (a) XRD for GO and GO-POSS. (b) Raman for GO and GO-POSS. (c) GO and GO-POSS dispersed in methanol solution. (d) AFM of GO and GO-POSS.

layer spacing of 0.87 nm. It demonstrates that GO-POSS has a more compact structure than GO, which was unexpected. Combined with the AFM of GO-POSS in Fig. 3(d), it can be found that a small amount of layer stacking appears in the grafted GO-POSS. It may be precisely because of the stacking of these layers causing the layer spacing to be shortened slightly, thus showing GO-POSS has a more compact structure. Theoretically speaking, an amide reaction occurs between the carboxyl group on GO and the amino group on POSS to remove one oxygen, yet, more oxygen is added from POSS, hence, it is a process of increasing oxygen content. It can be observed from Table 1 that the C/O ratio of GO is 2.47, the C/O ratio of GO-POSS is 2.42, and the O content after grafting has mildly increased, which echoes the above-mentioned theory. At the same time, in Fig. 3(c), GO (left) and GO-POSS (right) were dispersed in methanol at a concentration of 1 mg/mL, respectively, the color changes from the original brown to light black. This also corresponds to the XRD result above. In Fig. 3(b), there are typical characteristic peaks of carbon materials, the D peak (lattice defect of C atom) is about 1350 cm^{-1} , and the G peak (in-plane stretching vibration of C atom sp^2 hybridization) is about 1580 cm^{-1} . The I_D/I_G ratio of Raman is used to evaluate the defects of the carbon material, and a larger ratio means a larger defect in the crystal lattice. It is distinctly observed that the ratio of GO-POSS is obviously reduced compared to GO from 1.03 to 0.92, which is because the introduction of POSS reduces the disorder of GO. Thus, POSS not only makes GO more orderly and firm distribute in nanocomposite materials, but also greatly reduces its structural disorder, thereby greatly improving the role of GO in nanocomposite materials. Fig. 3(d) presents a typical AFM image for GO and GO-POSS. The thickness of the monolayer GO sheet is 1.17 nm, while the height of GO-POSS increases to 4.54 nm. It directly reveals that POSS was grafted successfully to the surface of GO.

To further confirm the success of the reaction, XPS spectra was introduced as shown in Fig. 4. Obviously, in Fig. 4(a), there are two strong peaks in GO and rGO corresponding to the C 1s (285.1 eV/290.1 eV) and O 1s (536.1 eV/533.1 eV) peaks, respectively. However, the survey scan result shows the additional peaks Si 2p peak and the N 1s peak in GO-POSS and rGO-POSS were located at 107.2 eV/103.1 eV and 407.1 eV/400.1 eV in GO-POSS and rGO-POSS, respectively. The C 1s and Si 2p high-resolution spectra were deconvoluted, and results are involved in Fig. 4(b–f). As shown in Fig. 4(b), the C 1s spectra of POSS were deconvoluted around 284.1 eV, 284.8 eV, 286.2 eV and 288.7 eV, corresponding to C–Si, C–C, C–NH₂⁺ and C–Si–O. Compared to the C1s of GO, the peaks of GO-POSS and rGO-POSS appears new peaks at 287.4 eV/288.1 eV (C–N) and 288.0 eV/290.6 eV (O=C–N). Meanwhile, it generated a slight variation in C–C (284.8 eV and 284.3 eV) and a sharply decrease in C=O (288.6 eV and 292.4 eV). Moreover, the C–N and O=C–N peaks generated in GO-POSS and rGO-POSS are not present in GO, indicating that a mass of POSS is grafted onto the surface of GO [32]. More powerfully evidence was presented in Fig. 3(f). It shows the high-resolution Si 2p spectra for rGO-POSS, with the appearance of Si–O–Si (102.4 eV) and Si–C (101.7 eV). Further data is shown in Table 1. The appearance of nitrogen and silicon also straightway proves that POSS has been resoundingly grafted to GO. The ratio of C to O increased sharply from 2.43 of GO-POSS to 5.38 of rGO-POSS. It thanks to the detachment of unstable oxygen-containing functional groups [40], especially –OH, on GO when the temperature rises above 200 °C.

The SEM cross-sectional images of the nanocomposites were used to

investigate the grafting and dispersion of rGO-POSS. The bright and sharp areas in the SEM images are attributed to the high conductivity of GO nanosheets [47], as shown in Fig. 5(b–f). There are only some bright areas in Fig. 5(a), which is due to the height difference of the fracture after brittle fracture. In Fig. 5(b–d), it can be expressly observed that the bright and sharp area of the cross-sectional fracture gradually increases without agglomeration, indicating that the rGO-POSS nanosheets are homogeneous dispersed in the matrix after grafting the PA6 long chain. Meanwhile, we could observe that the rGO-POSS is wrapped in space between the PA6 matrix (dark area). It is due to the in-situ polymerization method, in which rGO-POSS nanosheets are uniformly inserted into the CL monomer and enhances the interface bonding ability with the matrix, thereby providing the greatest possibility for the nanocomposites to obtain excellent properties. However, the sharp and bright area drops sharply starting from Fig. 5(e). It means that rGO-POSS has poor dispersion in the matrix and aggregates. The results showed that rGO-POSS nanosheets could not be homogeneous dispersed in the matrix due to agglomeration. After aggregation, it has a negative impact on the structure of the matrix, the stress transfer is severely hindered and the stress accumulation is gradually formed. As a result, the performance of the nanocomposites is deteriorated.

3.2. Mechanical performance and thermal stability of PA6/rGO-POSS nanocomposites

The effect of crystallization on polymer properties is more prominent [42], especially the mechanical properties and thermal properties. Generally, PA6 has three crystal forms: α crystal form, β crystal form and γ crystal form [48], in which the α crystal form is a thermodynamically stable crystal form and the β crystal form is an unstable crystal form [49–51]. PA6 is a polycrystalline polymer whose crystallites are in the form of α -type (α_1 phase) of $T \sim 212\text{--}214\text{ }^\circ\text{C}$, γ -type (γ -phase) of $T \sim 219\text{ }^\circ\text{C}$ and α -type (α_2 phase) of $T \sim 221\text{ }^\circ\text{C}$ [52]. On the basis of the DSC curve in Fig. 6(a) and T_m in Table 2, it is clear known that PA6 and its nanocomposite materials are thermodynamically stable α crystal form. It distinctly indicates that the GO-POSS added in the nanocomposites has little effect on crystallization. Herein, the increase of crystallinity will increase the stress, strength and hardness of the material while reducing the fracture elongation of the material, thereby making the material hard and brittle. The crystallinity of pure PA6 is 25.4%. With the addition of GO-POSS, the crystallinity of the PA6/rGO-POSS nanocomposites continues to increase. When 0.5 wt% and 0.9 wt% GO-POSS were added, its crystallinity corresponding to the 28.4% and 31.6%, and its elongation at break is significantly reduced, in Fig. 7(a). This is due to the increase in crystallinity, the PA6/rGO-POSS molecular chain is closely arranged, low porosity, strong intermolecular interaction and the chain motion becomes difficult, so that brittle fracture easily occurs. It can be observed from Fig. 6(b) that the thermal conductivity of the nanocomposites presents an overall trend of increasing first and then decreasing. With the addition of 0.3 wt% rGO-POSS, the thermal conductivity reached the maximum value of 0.30297, which is about 33.76% higher than pure PA6, and the heat dissipation performance of composite materials is the best. This is mainly due to the good dispersibility of 0.3 wt% rGO-POSS, which can effectively transfer heat. Conversely, when the loadings of rGO-POSS increase, the dispersibility of the particles becomes worse, and the heat cannot be effectively transferred, which leads to a decrease in thermal conductivity.

The mechanical performance of PA6/rGO-POSS nanocomposites with different loadings were illustrated in Fig. 7(a–d). Compared with pure PA6, in Fig. 7(a–b), it is obvious that 0.3 wt% PA6/rGO-POSS has the highest tensile strength and elongation at break, which are 71.75 MPa and 259.62% respectively. This enhanced effect is attributed to the uniform dispersion and strong adhesion of rGO-POSS, which can effectively transfer loads. Meanwhile, the 0.3 wt% PA6/rGO-POSS crystallinity is relatively low, increasing the toughness. In Fig. 7(c), it could be distinctly observed that Young's modulus of 0.1 wt% PA6/rGO-POSS is

Table 1
of element content of GO, rGO, GO-POSS, rGO-POSS calculated from XPS analysis.

Sample	C(%)	N(%)	O(%)	Si(%)
GO	71.21	–	28.79	–
rGO	86.69	–	13.31	–
GO-POSS	67.50	2.72	27.78	2.01
rGO-POSS	80.53	2.40	14.96	2.11

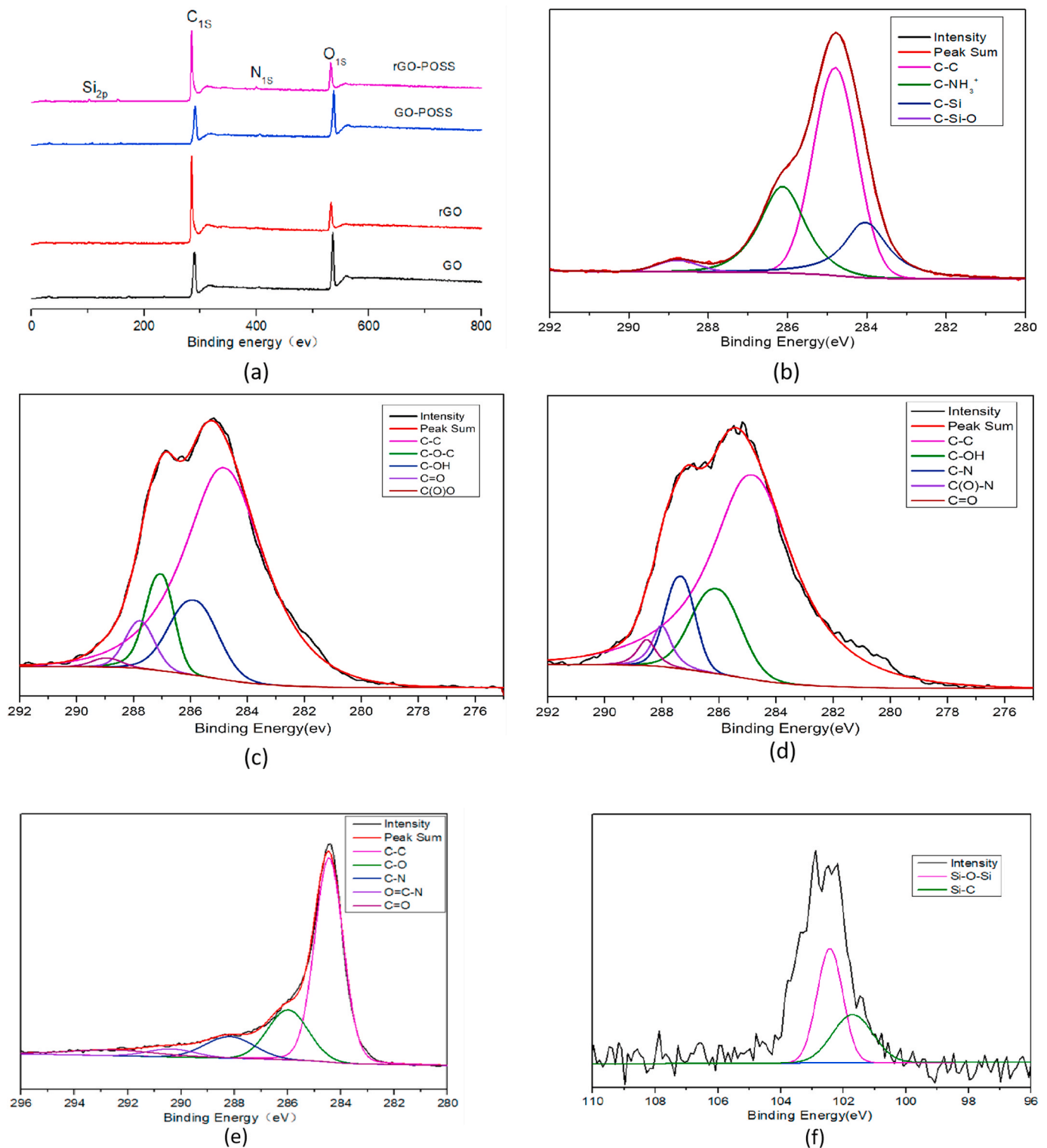


Fig. 4. (a) X-ray photoelectron spectra of POSS, GO, GO-POSS, rGO-POSS. (b) High-resolution XPS spectra of C 1s for POSS. (c–e) XPS spectra of C 1s for GO, GO-POSS and rGO-POSS. (f) XPS spectra of Si 2p for rGO-POSS.

higher than other content, which is 1658.22 MPa. The enhancement is mainly attributed to two situations: The first point is that POSS is a kind of compound composed of Si–O alternately connected silicon oxygen framework (cage-like structure). Its cage-like structure can play an elastic function similar to “marbles” and thus has a toughening effect on the polymer. The second point is that the octopus-shaped rGO-POSS is formed in PA6 and further expanded into a self-fixing structure with a

large mesh structure. It can be compared to be a bridge in a mesh structure, which can transfer the stress from the matrix to the rGO-POSS, hence, preventing the propagation of cracks and vastly improving the mechanical properties of the nanocomposites. When 0.5 wt% rGO-POSS was added, a remarkable decrease of mechanical properties occurs. It mainly thanks to the increasing of the rigidity of the molecular chain by addition of rigid particles rGO-POSS, which makes the ability of the

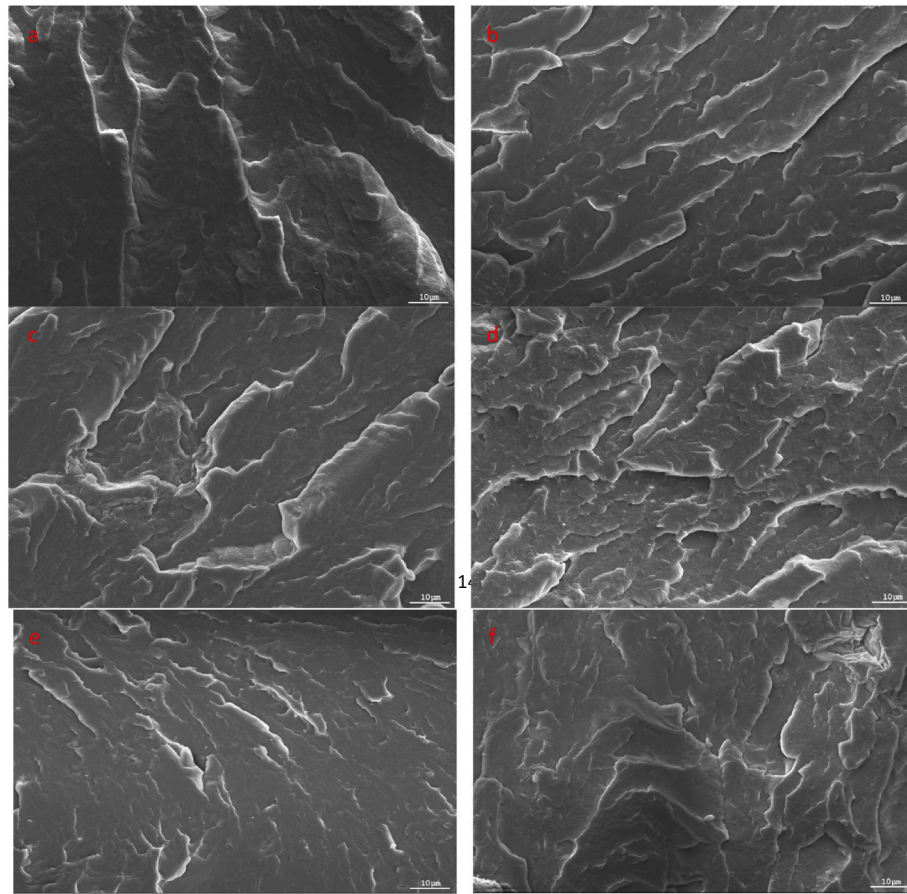


Fig. 5. SEM cross section of neat PA6 (a), 0.1 wt% PA6/rGO-POSS (b), 0.3 wt% PA6/rGO-POSS (c), 0.5 wt% PA6/rGO-POSS (d), 0.7 wt% PA6/rGO-POSS (e), 0.9 wt% PA6/rGO-POSS.

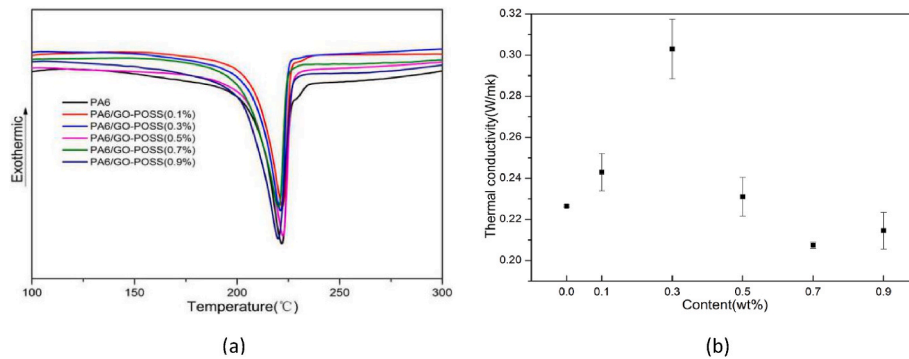


Fig. 6. (a) DSC curve of PA6 and its nanocomposites. (b) Thermal conductivity of PA6 and its nanocomposites.

Table 2

Main data of DSC test of PA6 and its nanocomposites.

Sample	$T_g(^{\circ}\text{C})$	$\Delta H_m(\text{J}\cdot\text{g}^{-1})$	$T_m(^{\circ}\text{C})$	X (%)
PA6	210.2	56.28	221.8	25.4
PA6/rGO-POSS(0.1 wt%)	209.7	55.36	222.0	25.0
PA6/rGO-POSS(0.3 wt%)	209.1	58.66	221.1	26.5
PA6/rGO-POSS(0.5 wt%)	209.1	62.91	222.4	28.4
PA6/rGO-POSS(0.7 wt%)	206.4	64.86	220.1	29.3
PA6/rGO-POSS(0.9 wt%)	206.7	69.99	220.0	31.6

molecular chain to resist deformation worse [25], and higher crystallinity limits the plastic deformation of the matrix. In Fig. 7(d), the Shore hardness first increased and then decreased with the increase of

rGO-POSS content. The Shore hardness value of 0.5 wt% PA6/rGO-POSS reaches the maximum value of 89.4, which is 23.62% higher than pure PA6. This is mainly due to the addition of rGO-POSS, which increases the rigidity of the molecular chain and thus increases the hardness of the PA6/rGO-POSS nanocomposites. However, as the content of rGO-POSS increases, rGO-POSS agglomerates in the matrix, which causes the rigidity of the molecular chain to weaken and the hardness decreases.

The fracture morphology of nanocomposites has undergone significant changes that can be distinctly observed from the brittle fracture morphology in Fig. 8. The fracture morphology of Fig. 8(a-c) is similar, with relatively flat fractures and slender filaments. However, Fig. 8(b-c) began to show uneven fractures. Addition of 0.1 wt% and 0.3 wt% of rGO-POSS has a partial effect on the matrix structure, but it can improve

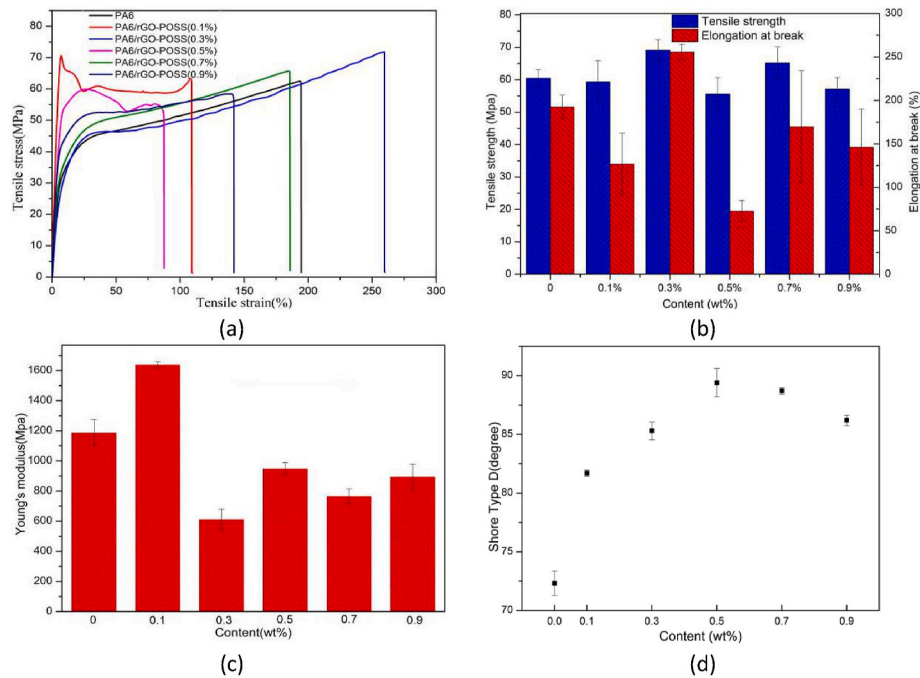


Fig. 7. (a) Typical stress-strain curves of pure PA6 and its nanocomposites. (b–c) Mechanical properties of PA6 nanocomposites. (d) Shore hardness of PA6/rGO-POSS composites with different contents.

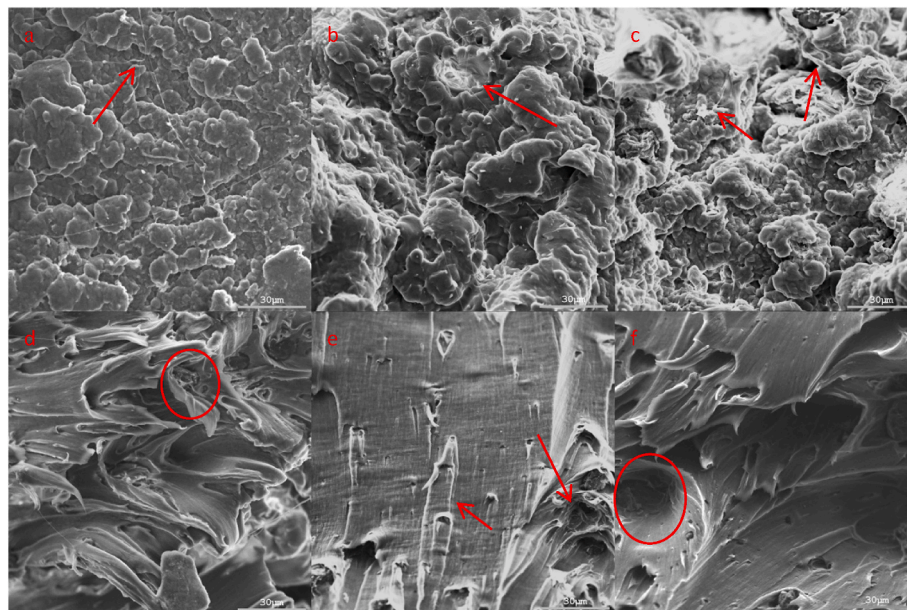


Fig. 8. Brittle fracture morphology of pure PA6 (a), 0.1 wt% PA6/rGO-POSS (b), 0.3 wt% PA6/rGO-POSS (c), 0.5 wt% PA6/rGO-POSS (d), 0.7 wt% PA6/rGO-POSS (e), 0.9 wt% PA6/rGO-POSS (f) after stretching.

the tensile strength and elongation at break of the nanocomposite to a certain extent. The hierarchical fiber fractures, “river-like” fractures and dimple fractures appeared with the addition of rGO-POSS continuing to increase. The fracture morphology of Fig. 8(d–f) is completely different from pure PA6, implicating that higher addition of rGO-POSS has a greater impact on the structure of the matrix. Combined with the SEM cross-sectional image, it can be seen that the higher loadings of rGO-POSS has poor dispersion in the matrix and is prone to agglomeration. This leads to the uneven stress of the nanocomposites during the stretching process, which is prone to brittle fracture and the main reason for the reduction of the tensile performance of the composite material.

3.3. Tribological properties of PA6 nanocomposites

To explore the tribological behavior of PA6/rGO-POSS nanocomposites, we recorded the curves of the relationship between the dynamic change of the friction coefficient and time, in Fig. 9(a). It could be clearly obtained from the polymer nanocomposites friction coefficient curves that there are two trends in friction behavior: the spurt period and stable period [53]. Here, in the initial stage of friction, the friction coefficient curve increased remarkably owing to a strong shearing action between the steel ball and protrusions of the sample. As the friction process continues, the friction coefficient begins to gradually

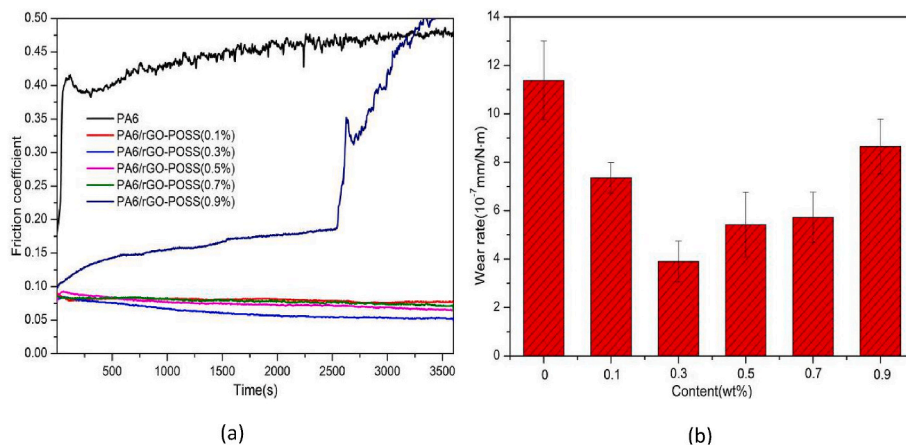


Fig. 9. (a) Friction coefficient of PA6/rGO-POSS nanocomposites with varying rGO-POSS content. (b) Variations of the wear rate of pure PA6 and PA6/rGO-POSS nanocomposites under different contents.

decrease and reach a stable stage, which is ascribed to the surfaces of both contact parts are rather smooth. The decrease later in friction coefficient may be because rGO-POSS could act as a solid lubricant. As the loading of rGO-POSS increases, the friction coefficient firstly exhibited a decrease, especially when 0.3 wt% rGO-POSS is added, the friction coefficient reached the lowest and then increased. Surprisingly, the friction coefficient increases steeply and forms a cliff when the additional amount is 0.9 wt%. It is mainly caused by the following situations: (1) the crystallinity is relatively higher, at the same time, the crystalline region-amorphous region formed inside the nanocomposite material accounts for a relatively larger proportion compared to other contents, which cannot transmit stress well and prevent cracks extension, (2) the interface bonding strength of rGO-POSS with matrix is reduced evidently, (3) a large amount of PA6/rGO-POSS was exposed to the outer surface, which cut and destroyed the integrity and smoothness of the transfer film with the steel ball, thereby leading to a sharply decrease in wear resistance. The average friction coefficient of PA6 and 0.1 wt%, 0.3 wt%, 0.5 wt%, 0.7 wt%, 0.9 wt% PA6/rGO-POSS was calculated to be 0.44, 0.08, 0.06, 0.07, 0.08, 0.24, respectively. The friction coefficient shows a trend of first decreasing and then increasing, which confirms the SEM images of the surface before and after friction. Fig. 9(b) reveals the wear rate of the nanocomposites under different loadings. The wear rate of 0.3 wt% PA6/rGO-POSS observably reduced by 65.72% compared to that of pure PA6. This is because a layer of wear-resistant transfer film was formed on the surface of the steel ball, which changes the wear form of the rigid ball and the block sample. Based on the above analysis, we could better determine that the remarkable enhancement in the friction and wear performance of the polymer is attributed to the self-fixation of octopus-shaped rGO-POSS in the matrix. Also, it enhances the interface bonding strength and prevents the initiation and propagation of cracks.

Fig. 10 reveals the microscopic morphology of the friction surface before and after the wear test to explore the wear mechanism. It could be clearly observed that there were a lot of dents, pits and cracks on the surface of pure PA6 samples before being worn, which precisely caused a serious matrix peeling off after abrasion. The wear surface of pure PA6 exhibits large-size abrasive particles, uneven folds and cracks extending inward marked by red arrows. More seriously, the uneven abrasive particles after peeling produce strong friction with the substrate and form grooves of different depths on the surface of the substrate, showing low wear resistance, meaning that the form of wear is mainly abrasive wear and adhesive wear. Compared with pure PA6, the surface with a loading of 0.1 wt% and 0.3 wt% still possess some microcracks, wrinkles and small-sized abrasive particles. Fortunately, due to the protective and lubricating effects of rGO-POSS on the matrix, these flaws are tightly bondage on the surface and cannot be extended to the surroundings,

thus obtaining higher wear resistance. Especially, when the loading is 0.3 wt%, the “octopus” structure formed by rGO-POSS is adequately fixed in the matrix, which can effectively transfer stress, significantly prevent the generation and expansion of cracks, vastly improve the interface bonding strength, and remarkably enhance the wear resistance of nanocomposite materials [10,28]. The abraded surface was relatively smooth and flat, with only a few small abrasive particles, indicating that the wear form transfers to abrasive wear. When the addition of rGO-POSS increased to 0.5 wt% and 0.7 wt%, obvious bumps are observed on the surface owing to rGO-POSS having poor dispersion at higher loading quantity and cannot be self-fixed in the matrix stably before wear. There are some pock-marked pits and small flakes on the worn surface, portending that the wear mechanism officially enters the stage of fatigue wear. However, when the loading of rGO-POSS increased to 0.9 wt%, the surface before abrasion emerged analogical and relatively fewer defects compared to pure PA6. A large number of irregular pock-marked pits and abrasive particles appeared on the worn surface, which implied that the interface bonding strength between rGO-POSS and the matrix distinctly decreased. Simultaneously, the form of wear became fatigue wear and abrasive wear. Fig. 11 demonstrates the 3D morphology and depth of the wear scar. As the loading of rGO-POSS increases, the depth and width of wear scars firstly exhibit a trend of decreasing, especially at the loading of 0.3 wt%, reaching the lowest value $3.27 \mu\text{m} \times 0.17 \text{ mm}$, and it subsequently increasing. The result presents that the addition of rGO-POSS enhances the interface bonding strength which was constituted by the reinforcement and the matrix, so that the nanocomposite material is harder to peel off during the friction process than pure PA6.

3.4. Analysis of wear mechanism

Based on the above analysis, the mechanism of friction and wear was exhibited in Fig. 12(a–b). Obviously, the hardness of steel balls is higher than that of polymers. Hence, there are a tiny gullies on the surface of the nanocomposites during the loading stress and sliding. As shown in Fig. 12(a), pure PA6 is subjected to cyclic reciprocating load stress during friction. The surface of the substrate is partially peeled off on account of some defects, thereby it exists some larger abrasive particles. The surface of the abraded substrate assumes a mass of wrinkles and cracks, and it possesses the characteristics of adhesive wear. The friction and wear mechanism of PA6 is mainly ascribed to abrasive wear and adhesive wear. Fig. 12(b) demonstrates the friction and wear mechanism of the nanocomposite with rGO-POSS added. Because of the “octopus” self-fixing structure formed between rGO-POSS and the matrix, it has strong interface adhesion. Meanwhile, the slightly peeled substrate and rGO-POSS will adhere to the surface of the steel ball

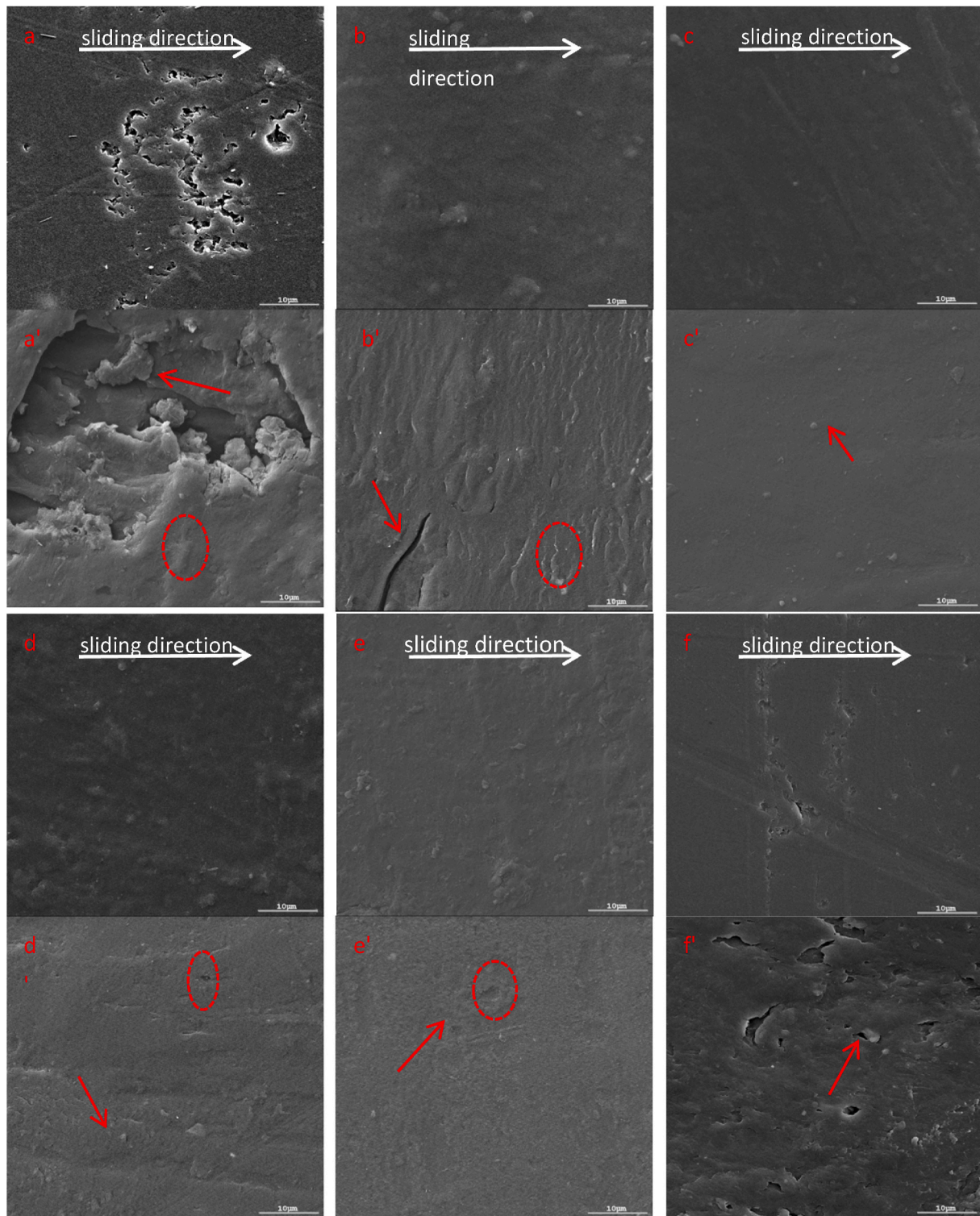


Fig. 10. SEM microstructure of the worn surfaces before and after friction conditions: a, b, c, d, e and f before wear test and a', b', c', d', e' and f' after wear test (the corresponding contents of a-f and a'-f' are 0 wt%, 0.1 wt%, 0.3 wt%, 0.5 wt%, 0.7 wt%, 0.9 wt% rGO-POSS).

during rapid cyclic reciprocating load stress, thereby forming a transfer film [28,54,55]. Besides, debris generated by friction will be alleviated, while the corrugation and cracks on the worn surface are reduced remarkably. Because of the strong interface bonding strength and the wear-resistant transfer film, the crack initiation and growth of the nanocomposites are seriously hindered. Besides, POSS nanoparticles can

effectively prevent the development of micro-crack tips and GO with high ductility and flexibility. When POSS is grafted onto GO, it can efficaciously buffer stress and suppress the generation and expansion of stress cracks, thereby improving the resistance of the nanocomposite material and reducing the coefficient of friction.

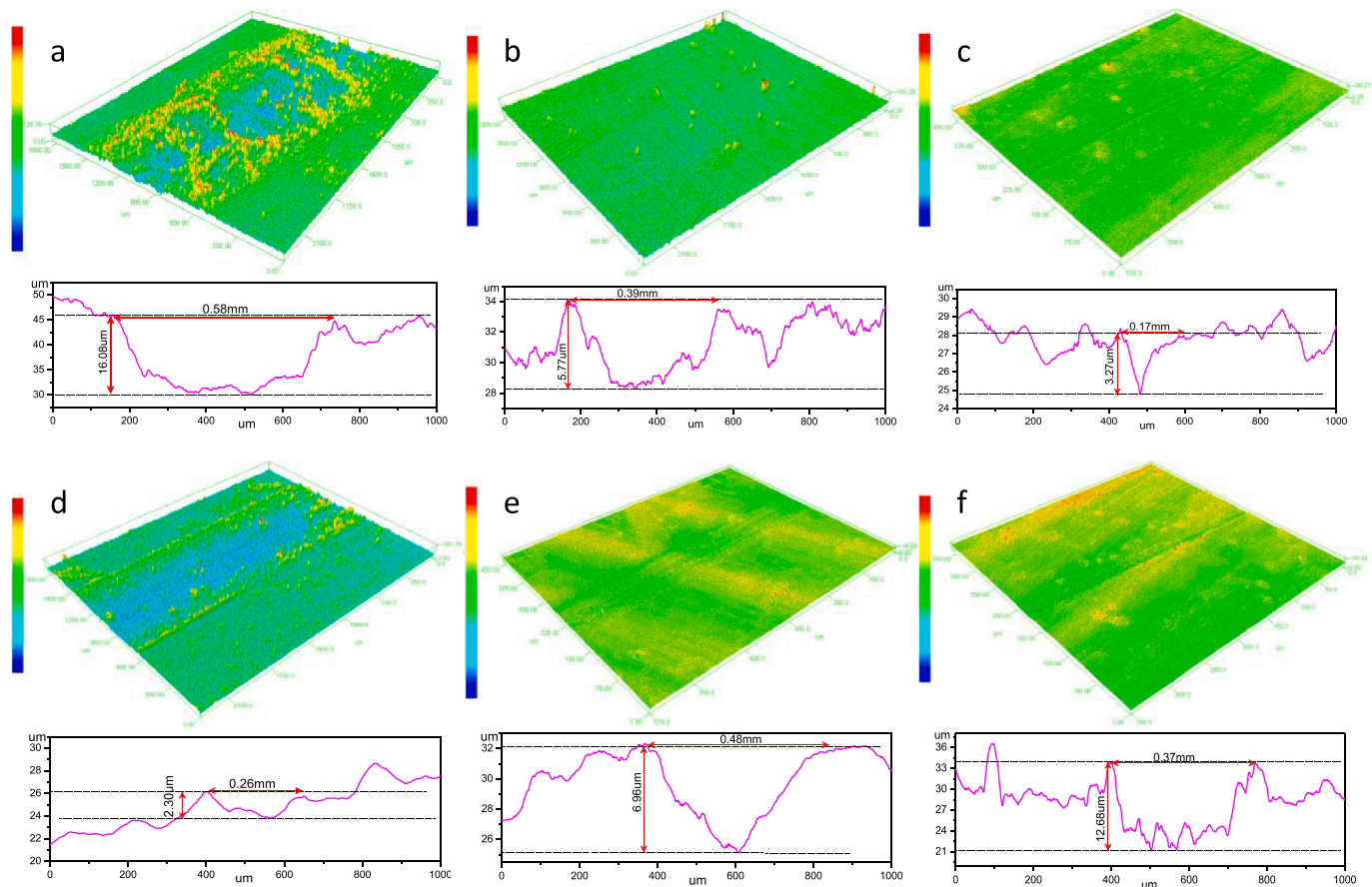


Fig. 11. 3D topographies of worn surface: a-f contents corresponding to 0% wt, 0.1 wt%, 0.3 wt%, 0.5 wt%, 0.7 wt%, 0.9 wt% rGO-POSS.

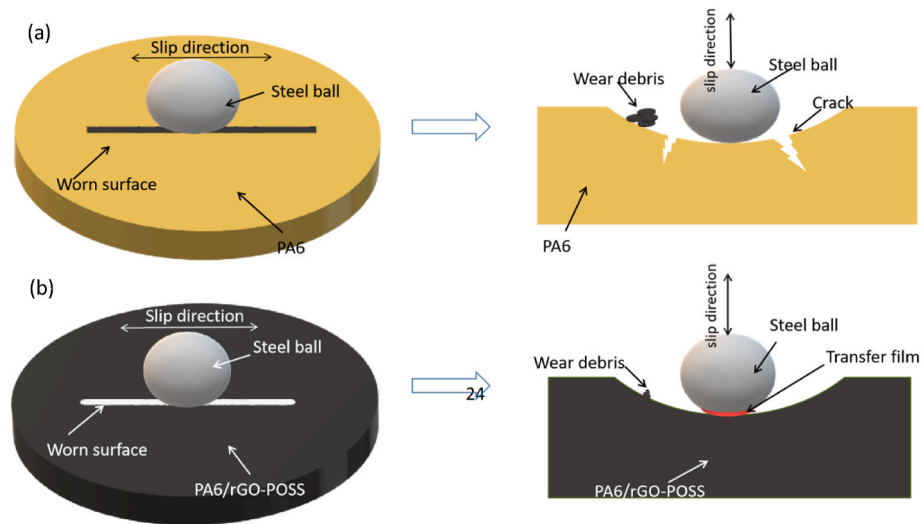


Fig. 12. Schematic diagram of wear mechanism: (a) PA6 and (b) PA6/rGO-POSS.

4. Conclusion

In this paper, the in-situ polymerization method was used to introduce multi-reaction sites GO-POSS with octopus-shaped structure, and a series of PA6/rGO-POSS nanocomposites with varying loadings of GO-POSS were synthesized. POSS shaped GO provides more reactive sites for grafting with PA6. The mechanical properties, stability, tribological

properties of nanocomposites were studied. The results demonstrate that the rGO-POSS can be self-fixed and equally dispersed in the matrix, keeping the matrix still being in the thermodynamically stable α crystal form, which improves the interfacial adhesion, accelerates stress transmitting and prevents the initiation and propagation of cracks. The strong interface bonding force between the matrix and rGO-POSS can form a protective layer with high-wearing feature on the surface of the steel ball

during the friction and wear process, which remarkably improves the tribological performance of the nanocomposites. Furthermore, compared with pure PA6, addition of only 0.3 wt% GO-POSS led to an approximately 86.36% reduction in friction coefficient from 0.44 to 0.06, and an increase in tensile strength and elongation at break by 14.98% and 33.90%, respectively. At the same time, the worn surface became smooth and flat with some small diameter abrasive particles. In addition, the depth and width of the wear scar after friction were measured and they were $3.27 \mu\text{m} \times 0.17 \text{ mm}$, which were reduced by 79.64% and 70.69%, respectively, compared to pure PA6. It further proves the strengthening effect of rGO-POSS on nanocomposite materials. This work supplies a high wear resistance PA6 nanocomposites in the field of engineering plastics.

CRediT authorship contribution statement

Lei Sun: Conceptualization, Methodology, Software, Formal analysis, Writing - original draft. **Ming Cao:** Data curation, Software, Methodology. **Fan Xiao:** Software, Writing - review & editing. **Jiaopeng Xu:** Software. **Yingbo Chen:** Conceptualization, Methodology, Resources, Data curation, Writing - review & editing, Project administration, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Sukumaran J, Pauw JD, Neis PD, Tóth LF, De Baets P. Revisiting polymer tribology for heavy duty application. *Wear* 2017;376–377:1321–32.
- [2] Álvarez-Láinez ML, Palacio RJA. Correlations between thermal and tensile behavior with friction coefficient in copolyamides 6/12. *Wear* 2017;372–373: 76–80.
- [3] Pereira AAC, Santos RR, d'Almeida JRM. Effects of incorporating titanate nanotubes on the mechanical properties of polyamide 11. *Polym Test* 2018;68: 238–47.
- [4] Unal H, Mimaroglu A. Friction and wear performance of polyamide 6 and graphite and wax polyamide 6 composites under dry sliding conditions. *Wear* 2012;289: 132–7.
- [5] Ambrósio JD, Balarim CVM, de Carvalho GB. Preparation, characterization, and mechanical/tribological properties of polyamide 11/Titanium dioxide nanocomposites. *Polym Compos* 2016;37:1415–24.
- [6] Shi KH, Ye L, Li GX. In situ stabilization of polyamide 6 with reactive antioxidant. *J Therm Anal Calorim* 2015;119:1747–57.
- [7] Voyer J, Jiang Y, Pakkanen TA, Diem A. Adhesive friction and wear of micro-pillared polymers in dry contact. *Polym Test* 2019;73:258–67.
- [8] Sathees Kumar S, Kanagaraj G. Investigation on mechanical and tribological behaviors of PA6 and graphite-reinforced PA6 polymer composites. *Arabian J Sci Eng* 2016;41:4347–57.
- [9] Karsli NG, Yilmaz T, Gul O. Effects of coupling agent addition on the adhesive wear, frictional and thermal properties of glass fiber-reinforced polyamide 6,6 composites. *Polym Bull* 2018;75:4429–44.
- [10] Karatas E, Gul O, Karsli NG, Yilmaz T. Synergetic effect of graphene nanoplatelet, carbon fiber and coupling agent addition on the tribological, mechanical and thermal properties of polyamide 6,6 composites. *Compos B Eng* 2019;163:730–9.
- [11] Morioka Y, Tsuchiya Y, Shioya M. Correlations between the abrasive wear, fatigue, and tensile properties of filler-dispersed polyamide 6. *Wear* 2015;338–339: 297–306.
- [12] Aghamohammadi H, Heidarpour A, Jamshidi R, Bayat O. Tribological behavior of epoxy composites filled with nanodiamond and Ti3AlC2 -TiC particles: a comparative study. *Ceram Int* 2019;45:9106–13.
- [13] Madhu HC, Edachery V, Lijesh KP, Perugu CS, Kailas SV. Fabrication of wear-resistant Ti3AlC2/Al3Ti hybrid aluminum composites by friction stir processing. *Metall Mater Trans* 2020;51:4086–99.
- [14] Jamshidi R, Heidarpour A, Aghamohammadi H, Eslami-Farsani R. Improvement in the mechanical and tribological behavior of epoxy matrix with the inclusion of synthesized Ti3AlC2 MAX particles. *J Compos Mater* 2019;53:3819–27.
- [15] Kazemi-Khasragh E, Bahari-Sambran F, Platzner C, Eslami-Farsani R. The synergistic effect of graphene nanoplatelets-montmorillonite hybrid system on tribological behavior of epoxy-based nanocomposites. *Tribol Int* 2020:106472.
- [16] Pan B, Li N, Chu G, Wei F, Liu J, Zhang J, et al. Tribological investigation of MC PA6 reinforced by boron nitride of single layer. *Tribol Lett* 2014;54:161–70.
- [17] Novoselov KS, Geim AK, Morozov SV, Jiang D, Zhang Y, Dubonos SV, et al. Electric field effect in atomically thin carbon films. *Science* 2004;306:666–9.
- [18] Potts JR, Dreyer DR, Bielawski CW, Ruoff RS. Graphene-based polymer nanocomposites. *Polymer* 2011;52:5–25.
- [19] Sengupta R, Bhattacharya M, Bandyopadhyay S, Bhowmick AK. A review on the mechanical and electrical properties of graphite and modified graphite reinforced polymer composites. *Prog Polym Sci* 2011;36:638–70.
- [20] Zhang X, Sun L, Chen Y, Yin C, Liu D, Ding E. Flame retarded polyamide-6 composites via in situ polymerization of caprolactam with perylene-3,4,9,10-tetracarboxylic dianhydride. *SN Applied Sciences* 2019;1.
- [21] Abbandanak SNH, Aghamohammadi H, Akbarzadeh E, Shabani N, Eslami-Farsani R, Kangoie M, et al. Morphological/SAXS/WAXS studies on the electrochemical synthesis of graphene nanoplatelets. *Ceram Int* 2019;45: 20882–90.
- [22] Chen J, Yao B, Li C, Shi G. An improved Hummers method for eco-friendly synthesis of graphene oxide. *Carbon* 2013;64:225–9.
- [23] Chen J, Li Y, Huang L, Li C, Shi G. High-yield preparation of graphene oxide from small graphite flakes via an improved Hummers method with a simple purification process. *Carbon* 2015;81:826–34.
- [24] Kim J, Cote LJ, Kim F, Yuan W, Shull KR, Huang JX. Graphene oxide sheets at interfaces. *J Am Chem Soc* 2010;132:8180–6.
- [25] Chen Y, Li D, Yang W, Xiao C. Enhancement of mechanical, thermal and tribological properties of AAPS-modified graphene oxide/polyamide 6 nanocomposites. *Compos B Eng* 2018;138:55–65.
- [26] Lee SH, Kim HW, Hwang JO, Lee WJ, Kwon J, Bielawski CW, et al. Three-dimensional self-assembly of graphene oxide platelets into mechanically flexible macroporous carbon films. *Angew Chem Int Ed Engl* 2010;49:10084–8.
- [27] Lee SH, Dreyer DR, An J, Velamakanni A, Piner RD, Park S, et al. Polymer brushes via controlled, surface-initiated atom transfer radical polymerization (ATRP) from graphene oxide. *Macromol Rapid Commun* 2010;31:281–8.
- [28] Wang J, Zhou S, Huang J, Zhao G, Liu Y. Interfacial modification of basalt fiber filling composites with graphene oxide and polydopamine for enhanced mechanical and tribological properties. *RSC Adv* 2018;8:12222–31.
- [29] Cordes DB, Lickiss PD, Rataboul F. Recent developments in the Chemistry of cubic polyhedral oligosilsesquioxanes. *Chem Rev* 2010;110:2081–173.
- [30] Tanaka K, Chujo Y. Advanced functional materials based on polyhedral oligomeric silsesquioxane (POSS). *J Mater Chem* 2012;22:1733–46.
- [31] Wu S, Hayakawa T, Kakimoto M, Oikawa H. Synthesis and characterization of organo soluble aromatic polyimides containing POSS in main chain derived from double-decker-shaped silsesquioxane. *Macromolecules* 2008;41:3481–7.
- [32] Yuan R, Ji L, Wu Y, Li H, Ju P, Chen L, et al. "Plate-anchor" shaped POSS-functionalized graphene oxide with self-fixing effect in polyimide matrix: molecular dynamic simulations and experimental analysis. *Compos Sci Technol* 2019;176:103–10.
- [33] Li GZ, Wang LC, Ni HL, Pittman CU. Polyhedral oligomeric silsesquioxane (POSS) polymers and copolymers: a review. *J Inorg Organomet Polym Mater* 2001;11: 123–54.
- [34] Schwab JJ, Lichtenhan JD. Polyhedral oligomeric silsesquioxane (POSS)-based polymers. *Appl Organomet Chem* 1998;12:707–13.
- [35] Liao Q, Zhang Q, Wang X, Li X, Deng G, Meng Z, et al. Facile fabrication of POSS-Modified MoS2/PMMA nanocomposites with enhanced thermal, mechanical and optical limiting properties. *Compos Sci Technol* 2018;165:388–96.
- [36] Zhou Q, Cong Y, Wu N, Loo LS. The microstructure of polyamide 6 and polyamide 6/polyhedral oligomeric silsesquioxane nanocomposites synthesized by phase inversion procedure under electric field. *Appl Surf Sci* 2015;357:1454–62.
- [37] Phillips SH, Haddad TS, Tomczak SJ. Developments in nanoscience: polyhedral silsesquioxane (POSS)-polymers oligomeric. *Curr Opin Solid State Mater Sci* 2004; 8:21–9.
- [38] Hussain H, Shah SM. Recent developments in nanostructured polyhedral oligomeric silsesquioxane-based materials via 'controlled' radical polymerization. *Polym Int* 2014;63:835–47.
- [39] Wu J, Mather PT. POSS polymers: physical properties and biomaterials applications. *Polym Rev* 2009;49:25–63.
- [40] Xu Z, Gao C. In situ Polymerization approach to graphene-reinforced nylon-6 composites. *Macromolecules* 2010;43:6716–23.
- [41] Zhao C, Hu G, Justice R, Schaefer DW, Zhang S, Yang M, et al. Synthesis and characterization of multi-walled carbon nanotubes reinforced polyamide 6 via in situ polymerization. *Polymer* 2005;46:5125–32.
- [42] Ding P, Su S, Song N, Tang S, Liu Y, Shi L. Highly thermal conductive composites with polyamide-6 covalently-grafted graphene by an in situ polymerization and thermal reduction process. *Carbon* 2014;66:576–84.
- [43] Liu Y, Fu X, Duan J, Yang G. Self-assembly morphology evolution of the polyamide 6 (PA6) component in the PA6/polyethylene glycol system by in situ polymerization of ϵ -caprolactam monomer. *Polym Int* 2018;67:874–82.
- [44] Ramasundaram SP, Kim KJ. In-situ synthesis and characterization of polyamide 6/ POSS nanocomposites. *Macromol Symp* 2007;249–250:295–302.
- [45] Li YF, Zhu JH, Wei SY, Ryu J, Sun LY, Guo ZH. Poly(propylene)/Graphene nanoplatelet nanocomposites: melt rheological behavior and thermal, electrical, and electronic properties. *Macromol Chem Phys* 2011;212:1951–9.
- [46] Xiao F, Hu XY, Chen YB, Zhang YF. Porous Zr-based metal-organic frameworks (Zr-MOFs)-Incorporated thin-film nanocomposite membrane toward enhanced desalination performance. *ACS Appl Mater Interfaces* 2019;11:47390–403.
- [47] Xu Z, Gao C. In situ polymerization approach to graphene-reinforced nylon-6 composites. *Macromolecules* 2010;43:6716–23.
- [48] Fomes TD, Paul DR. Crystallization behavior of nylon 6 nanocomposites. *Polymer* 2003;44:3945–61.
- [49] Li YY, Goddard WA. Nylon 6 crystal structures, folds, and lamellae from theory. *Macromolecules* 2002;35:8440–55.

- [50] Medellin-Rodriguez FJ, Larios-Lopez L, Zapata-Espinoza A, Davalos-Montoya O, Phillips PJ, Lin JS. Melting behavior of polymorphs: molecular weight dependence and steplike mechanisms in nylon-6. *Macromolecules* 2004;37: 1799–809.
- [51] Penel-Pierron L, Depecker C, Seguela R, Lefebvre JM. Structural and mechanical behavior of nylon 6 films part I. Identification and stability of the crystalline phases. *J Polym Sci B Polym Phys* 2001;39:484–95.
- [52] Scaffaro R, Maio A. Optimization of two-step techniques engineered for the preparation of polyamide 6 graphene oxide nanocomposites. *Compos B Eng* 2019; 165:55–64.
- [53] Li C, Xiang M, Ye L. Intercalation structure and highly enhancing tribological performance of monomer casting nylon-6/graphene nano-composites. *Compos Appl Sci Manuf* 2017;95:274–85.
- [54] Ran J, Xie H, Lai X, Li H, Zeng X. Significant improvement of tribological performances of polyamide 46/polyphenylene oxide alloy by functionalized zirconium phosphate. *Tribol Int* 2018;128:204–13.
- [55] Palabiyik M, Bahadur S. Tribological studies of polyamide 6 and high-density polyethylene blends filled with PTFE and copper oxide and reinforced with short glass fibers. *Wear* 2002;253:369–76.